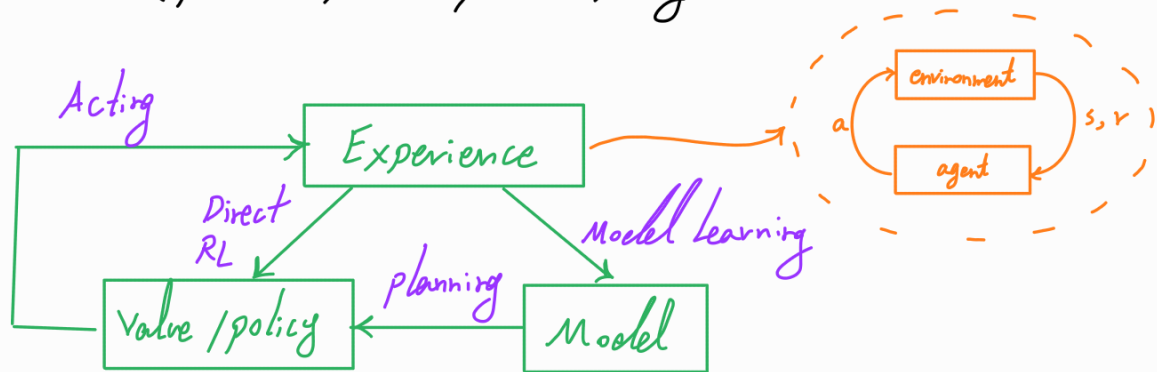


Model-based RL

In the settings with unknown environment model (transition function P and reward function R), model-based approaches aim to

- learn (estimate) the underlying model, and
- learn (plan for) an optimal policy over the learned model.



Dyna-Q algorithm: An algorithm that applies one-step interaction with the environment and K step planning.

Note: If the transition function is deterministic, then a single visit of state-action pair (s, a) is enough to learn $P(s' | s, a)$. Similarly, if the reward function is deterministic, then a single visit of state-action pair (s, a) is enough to learn $R(s, a)$.

The diagram shows a state-action pair $s=2, a=1$ leading to a next state $s'=4$ and a reward $r=-1$ via a transition function P . A policy π is also shown. The text below the diagram states: $|S||A|$ samples to learn P and R .

$$\left\{ \begin{array}{l} P(s'=1 | s=2, a=1) = 0 \\ \vdots \\ P(s'=4 | s=2, a=1) = 1, \quad R(s=2, a=1) = -1 \\ \vdots \\ P(s'=n | s=2, a=1) = 0 \end{array} \right.$$

Dyna-Q

Iteratively perform
• 1 step direct RL
• K steps planning
 $Q^* \rightarrow \pi^*$

Model-based method

- Initialize $\hat{Q}(s, a)$
- Initialize $\hat{P}(\cdot | s, a)$ and $\hat{R}(s, a)$
- Iterate

Interaction / experience

- $s \leftarrow$ current state
- Take action a using an ϵ -greedy policy over $\hat{Q}$ at s ;
observe reward $R(s, a)$ and next state s'

Model learning • Update $\hat{P}(\cdot | s, a)$ and $\hat{R}(s, a)$

Direct RL • $\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + \alpha [R(s, a) + \gamma \max_{a' \in A} \hat{Q}(s', a') - \hat{Q}(s, a)]$

planning

- Iterate K times
 - $s \leftarrow$ random previously observed state
 - $a \leftarrow$ random previously taken action at s
 - sample s' from $\hat{P}(\cdot | s, a)$ and consider $\hat{R}(s, a)$
 - $\hat{Q}(s, a) \leftarrow \hat{Q}(s, a) + \alpha [\hat{R}(s, a) + \gamma \max_{a' \in A} \hat{Q}(s', a') - \hat{Q}(s, a)]$

Value Iteration

• Based on V : $V_{t+1}(s) = \max_{a \in A} [R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V_t(s')]]$

• Based on Q : $Q_{t+1}(s) = R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [\max_{a' \in A} Q_t(s', a')]$

P unknown